Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 8 Handout

Tuesday, October 6

Problem 8.1

For each distribution below, if a tree graphical model exists that corresponds to the distribution, construct one. If not, explain.

(a) $p_{W,X,Y,Z} = p_W \, p_{X|W} \, p_{Y|X} \, p_{Z|X}$

(b) $p_{X,Y,Z} = p_X \, p_Y \, p_{Z|X,Y}$

Problem 8.2

Consider binary random variables $X_1, X_2, X_3 \in \{0, 1\}$ that form a Markov chain $X_1 \leftrightarrow X_2 \leftrightarrow X_3$ with

$$p_{X_2}(x_2) = 1/2, \quad x_2 = 0, 1$$

and

$$p_{X_1|X_2}(x_1|x_2) = \begin{cases} 5/6 & x_1 = x_2 \\ 1/6 & x_1 \neq x_2 \end{cases} \quad \text{and} \quad p_{X_3|X_2}(x_3|x_2) = \begin{cases} 1 & x_3 \neq x_2 \\ 0 & x_3 = x_2. \end{cases}$$

(a) We seek a tree graphical model with uniform node potentials that represents the joint distribution p_{X_1,X_2,X_3}. Define the tree structure and specify the corresponding edge potentials.

(b) Express p_{X_1,X_3} in terms of the potential functions defined in part (a).

(c) Determine $p_{X_3|X_1}(0|0)$.

(d) Construct the tree graphical model and corresponding potentials given $X_1 = 0$. Compute $p_{X_3|X_1}(0|0)$ using the updated potentials.